

MATHEMATICAL ANALYSIS 2

Seminar 9 – supplementary exercises

Solved exercises

 **Problem 1.** Compute $\int_0^a \frac{\ln(1+ax)}{1+x^2} dx$, $a \geq 0$.

 **Solution.** Denote the integral by $I(a)$. In order to find $I(a)$, we compute $I'(a)$ first:

$$I'(a) = \frac{\ln(1+a^2)}{1+a^2} + \int_0^a \frac{\partial}{\partial a} \left(\frac{\ln(1+ax)}{1+x^2} \right) dx = \frac{\ln(1+a^2)}{1+a^2} + \int_0^a \frac{x}{(1+ax)(1+x^2)} dx.$$

Then

$$\frac{x}{(1+ax)(1+x^2)} = \frac{1}{a^2+1} \left(-\frac{a}{1+ax} + \frac{x+a}{1+x^2} \right),$$

so

$$\begin{aligned} \int_0^a \frac{x}{(1+ax)(1+x^2)} dx &= \frac{1}{1+a^2} \left(-\int_0^a \frac{a}{1+ax} dx + \int_0^a \frac{x}{1+x^2} dx + \int_0^a \frac{a}{1+x^2} dx \right) \\ &= \frac{1}{1+a^2} \left(-\ln(1+ax) + \frac{1}{2} \ln(1+x^2) + a \cdot \arctg x \right) \Big|_{x=0}^{x=a} \\ &= \frac{1}{1+a^2} \left(-\ln(1+a^2) + \frac{1}{2} \ln(1+a^2) + a \cdot \arctg a \right) \\ &= 1+a^2 \left(-\frac{1}{2} \ln(1+a^2) + a \cdot \arctg a \right). \end{aligned}$$

It follows that

$$\begin{aligned} I'(a) &= \frac{\ln(1+a^2)}{1+a^2} + \frac{1}{1+a^2} \left(-\frac{1}{2} \ln(1+a^2) + a \cdot \arctg a \right) = \frac{1}{1+a^2} \left(\frac{1}{2} \ln(1+a^2) + a \cdot \arctg a \right) \\ &= \frac{1}{2} \left(\ln(1+a^2) \cdot \arctg' a + (\ln(1+a^2))' \cdot \arctg a \right) = \frac{1}{2} (\ln(1+a^2) \cdot \arctg a)', \end{aligned}$$

hence

$$I(a) = \frac{1}{2} \ln(1+a^2) \cdot \arctg a + c,$$

where

$$0 = I(0) = c.$$

Concluding,

$$I(a) = \frac{1}{2} \ln(1+a^2) \cdot \arctg a, \quad a \geq 0.$$



 **Problem 2.** Compute $\int_0^\infty \frac{\arctg 2x^2 - \arctg x^2}{x^2} dx$.

 **Solution.** Let

$$I(a) = \int_0^\infty \frac{\arctg ax^2}{x^2} dx \quad (a > 0).$$

The integral is convergent, since

$$\lim_{x \rightarrow 0} \frac{\operatorname{arctg} ax^2}{x^2} = a \quad (\text{apply [C2, Seminar 6] with } p = 0 < 1)$$

$$\lim_{x \rightarrow \infty} x^2 \cdot \frac{\operatorname{arctg} ax^2}{x^2} = \frac{\pi}{2} \quad (\text{apply [C1, Seminar 6] with } p = 2 > 1).$$

Then

$$I'(a) = \int_0^\infty \frac{1}{1+a^2x^4} dx$$

$$y = a^2x^4, \quad x = a^{-\frac{1}{2}}y^{\frac{1}{4}}, \quad dx = \frac{1}{4}a^{-\frac{1}{2}}y^{-\frac{3}{4}} dy$$

$$I'(a) = \frac{1}{4}a^{-\frac{1}{2}} \int_0^\infty \frac{y^{-\frac{3}{4}}}{1+y} dy = \frac{1}{4}a^{-\frac{1}{2}} \beta\left(\frac{1}{4}, \frac{3}{4}\right) = \frac{1}{4}a^{-\frac{1}{2}} \frac{\pi}{\sin \frac{\pi}{4}} = \frac{\pi\sqrt{2}}{4} a^{-\frac{1}{2}}.$$

Concluding,

$$\int_0^\infty \frac{\operatorname{arctg} 2x^2 - \operatorname{arctg} x^2}{x^2} dx = I(2) - I(1) = \int_1^2 I'(a) da = \frac{\pi\sqrt{2}}{4} \int_1^2 a^{-\frac{1}{2}} da$$

$$= \frac{\pi\sqrt{2}}{4} \cdot 2a^{\frac{1}{2}} \Big|_1^2 = \frac{\pi}{2} \sqrt{2a} \Big|_1^2 = \pi \left(1 - \frac{\sqrt{2}}{2}\right).$$



Problem 3. Let $F(t) = \int_0^1 \varphi(t, x) f(x) dx$, where $f \in C([0, 1])$ and

$$\varphi : [0, 1] \times [0, 1] \rightarrow \mathbb{R}, \quad \varphi(t, x) = \begin{cases} t(1-x), & 0 \leq t \leq x \leq 1 \\ (1-t)x, & 0 \leq x \leq t \leq 1 \end{cases}.$$

Compute F' and F'' .

Solution. Since

$$F(t) = \int_0^t \underbrace{\varphi(t, x)}_{x \leq t} f(x) dx + \int_t^1 \underbrace{\varphi(t, x)}_{x \geq t} f(x) dx = \int_0^t (1-t)x f(x) dx + \int_t^1 t(1-x) f(x) dx,$$

it follows that

$$F'(t) = \left[(1-t)t f(t) + \int_0^t (-x f(x)) dx \right] + \left[-t(1-t) f(t) + \int_t^1 (1-x) f(x) dx \right]$$

$$= -\int_0^t x f(x) dx + \int_t^1 (1-x) f(x) dx = -\int_0^t x f(x) dx + \int_t^1 f(x) dx - \int_t^1 x f(x) dx$$

$$= \int_t^1 f(x) dx - \int_0^1 x f(x) dx.$$

Moreover,

$$F''(t) = -f(t).$$



Problem 4. Find $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$, where $g(x, y) = \int_{x^2+y^2}^{x^3+y^3} f(2x^2+y^2+t) dt$.

Solution. Rewrite g , using the change of variable $u = 2x^2 + y^2 + t$, $du = dt$. Then

$$g(x, y) = \int_{3x^2+2y^2}^{x^3+2x^2+y^3+y^2} f(u) du,$$

$$\begin{aligned}\frac{\partial g}{\partial x} &= \frac{\partial (x^3 + 2x^2 + y^3 + y^2)}{\partial x} f(x^3 + 2x^2 + y^3 + y^2) - \frac{\partial (3x^2 + 2y^2)}{\partial x} f(3x^2 + 2y^2) \\ &= (3x^2 + 4x) f(x^3 + 2x^2 + y^3 + y^2) - 6x \cdot f(3x^2 + 2y^2)\end{aligned}$$

and

$$\begin{aligned}\frac{\partial g}{\partial y} &= \frac{\partial (x^3 + 2x^2 + y^3 + y^2)}{\partial y} f(x^3 + 2x^2 + y^3 + y^2) - \frac{\partial (3x^2 + 2y^2)}{\partial y} f(3x^2 + 2y^2) \\ &= (3y^2 + 2y) f(x^3 + 2x^2 + y^3 + y^2) - 4y \cdot f(3x^2 + 2y^2).\end{aligned}$$



Problems for self study

Compute:

1. $\int_0^\infty \frac{\operatorname{arctg} 3x - \operatorname{arctg} x}{x} dx$
2. $\int_0^\infty \frac{\ln(1 + a^2 x^2)}{1 + x^2} dx \quad (a \geq 0)$
3. $\int_0^\infty e^{-x^2} \cos ax dx \quad (a \geq 0)$
4. $h'(x)$, with $h(x) = \int_{x^2}^{2x^2+1} f(3x+t) dt$
5. $f'(y)$, with $f(y) = \int_{a+y}^{b+y} \frac{\sin^2(xy)}{x} dx$

Answers to the problems for self study

1. $\frac{\pi \ln 3}{2}$
2. $\pi \ln(a+1)$
3. $\frac{\sqrt{\pi}}{2} e^{-\frac{a^2}{4}}$
4. $(4x+3)f(2x^2+3x+1) - (2x+3)f(x^2+3x)$
5. $\frac{2y+b}{y^2+by} \sin(y^2+by) - \frac{2y+a}{y^2+ay} \sin(y^2+ay)$